

Senior Ontology - Smart Route & User Management API

Unit test specification: Find a route with no pollution preference (3.0)

Scope note: Backend tests target server.py (FastAPI). DFD 3.1–3.4: route geometry from Google Map API is represented as encoded polyline + metrics; POST /scoreRoutes scores routes with focus_pollutants None. Pollution API and Relative Humidity API are mocked. Flutter Directions / route list UI is out of scope.

Test suite — Process 3.0 Find a route with no pollution preference

Test Case ID	TC-ROUTE-3.0
Python module	UnitTest_3_0_Route_No_Pollution_Preference.py
Priority	High
Data store	N/A (stateless scoring); optional ontology files patched in tests

DFD sub-processes covered

3.1 Get route input — Encoded polyline and route metrics; decode_poly, sample points, cache key. Tests: test_decode_poly_and_sample_points_and_cache_key; fixture in dfd_route_fixtures.py.

3.2 Generate route list — Client builds list; server scores each candidate. Integration: test_dfd_3_0_score_routes_without_focus_pollutants (focus_pollutants None).

3.3–3.4 Selected route detail — risk_score, di, dt, dp, dw via evaluate_route and helpers. Tests: test_evaluate_route_all_branches, test_route_pollution_score_*, test_critic_weights_and_weather_score_branches.

External APIs (mocked) — Pollution: test_fetch_aqi_cache_hit_and_exception_branch, test_extract_pollutants_missing_concentration_and_alias. Humidity: test_fetch_humidity_branches. Ontology: test_ontology_explain_and_adjustment_branches, test_get_ontology_maps_missing_file_branch.

Step	Action	Expected Result	Status
1	POST /scoreRoutes with focus_pollutants None	scored routes with risk_score	Pass
2	Mocked fetch_aqi + fetch_humidity + decode_poly	fixture data along sampled points	Pass
3	Supporting unit tests (scoring + APIs + ontology)	all branches Pass (see DFD text)	Pass

DFD / requirement	Unit test method
3.1 Polyline / sampling	test_decode_poly_and_sample_points_and_cache_key
3.0 Integration no preference	test_dfd_3_0_score_routes_without_focus_pollutants
3.2–3.4 Scoring + detail	test_route_pollution_score_*, test_evaluate_route_all_branches, test_critic_weights_and_weather_score_branches
Pollution API slice	test_fetch_aqi_*, test_extract_pollutants_*
Humidity API slice	test_fetch_humidity_branches
Ontology DSS	test_ontology_explain_and_adjustment_branches, test_get_ontology_maps_missing_file_branch

Fixture: backend_tests/dfd_route_fixtures.py (shared with 4.0).

Coverage: run backend_tests/run_coverage.sh — UnitTest_3_0_Route_No_Pollution_Preference.py 100%. Generated 2026-04-06 10:53.